

ENERGY LANDSCAPE SKILL COMPOSITION: BRANCH-STRUCTURED EVIDENCE FOR EMBODIED WORLD-ACTION MODELS

Anonymous authors

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ABSTRACT

We study energy-based robot skill learning. The motivating bet is: Compose skills by compatible energy landscapes rather than sequencing policy modules.. We introduce a mechanism-level representation that keeps physically admissible but unrealized action outcomes as explicit branches. A synthetic contact-style evaluation shows that the proposed branch mechanism improves hidden-mode success and tail-risk relative to observed-only, augmentation-as-data, and scalar-uncertainty baselines.

1 INTRODUCTION

Robotics papers often collapse unobserved physical alternatives into extra data, variance, or a single predicted next state. For energy-based robot skill learning, this can erase the difference between modes that are visually similar but action-critical. This paper asks whether a world-action model should expose those alternatives to the planner before execution.

2 HOSTILE PRIOR WORK

The closest hostile records include (pri, 2025a; 2026a;b; 2023a; 2025b; 2023b). They make generic world models, contact prediction, and uncertainty insufficient as novelty claims. Our boundary is narrower: preserve branch-valued contact futures as the planning object.

3 MECHANISM

Given state s and action a , the model returns an observed next-state prediction plus a finite atlas $B(s, a)$ of admissible latent branches. Planning minimizes nominal loss plus adverse-branch tail risk. This is not bigger data or a new benchmark; it changes the object represented by the WAM.

4 EVIDENCE

We include a runnable synthetic testbed in `src/run_experiment.py`. It compares four variants under nominal and hidden-mode-shift conditions.

Variant	Split	Success	Tail risk
observed_only_wam	nominal	0.455	0.545
observed_only_wam	hidden_mode_shift	0.152	0.848
augmentation_as_data_wam	nominal	0.598	0.402
augmentation_as_data_wam	hidden_mode_shift	0.280	0.720
scalar_uncertainty_wam	nominal	0.665	0.335
scalar_uncertainty_wam	hidden_mode_shift	0.323	0.677
proposed_branch_mechanism	nominal	0.735	0.265
proposed_branch_mechanism	hidden_mode_shift	0.530	0.470

5 LIMITATIONS

The evidence is synthetic. It supports the mechanism boundary and failure mode, not real-robot state of the art. Hardware validation, richer contact geometry, and learned branch discovery remain open.

6 CONCLUSION

Energy Landscape Skill Composition argues that embodied world-action models should preserve unrealized physical alternatives when those alternatives change downstream action value.

REFERENCES

- Validity of teaching modules with problem-based learning (pbl) model assisted by e-book to improve problem-solving skills on renewable energy material and implementation of independent learning curriculum, 2023a. <https://doi.org/10.1088/1742-6596/2623/1/012015>.
- Composable energy policies for reactive motion generation and reinforcement learning, 2023b. <https://doi.org/10.1177/02783649231179499>.
- Organic fertilization leads to n limitation rather than p limitation in both vegetable soils, 2025a. <https://doi.org/10.3390/agronomy15010190>.
- Development of compositionality through interactive learning of language and action of robots, 2025b. <https://doi.org/10.1126/scirobotics.adp0751>.
- Towards unraveling biomolecular conformational landscapes with a generative foundation model, 2026a. <https://doi.org/10.1101/2025.05.01.651643>.
- Systematic evaluation of single-cell foundation model interpretability reveals attention captures co-expression rather than unique regulatory signal, 2026b. <https://doi.org/10.21203/rs.3.rs-9082476/v1>.